

Automated Screening of Parkinson’s Disease: Design, Development, and Artificial Intelligence Using Non-Contact Sensing

Abstract

Parkinson’s disease is a progressive neurodegenerative disorder whose motor symptoms are reflected in measurable alterations in gait and human movement. Early and reliable screening is essential for improving patient outcomes; however, conventional clinical assessments often rely on subjective evaluation by specialists and are typically limited to controlled clinical environments.

Non-contact sensing technologies provide a promising alternative by enabling unobtrusive and privacy-preserving monitoring of human motion without requiring wearable devices or physical interaction with the subject. When combined with modern artificial intelligence techniques, these sensing systems have the potential to support automated and scalable screening tools.

This work explores the design and development of an artificial intelligence-based framework for the automated screening of Parkinson’s disease using signals acquired from non-contact sensing devices. The proposed methodology considers the full processing pipeline, including signal acquisition, signal processing, time-frequency representation, and machine learning-based classification. Motion-related information captured by multiple sensor nodes is transformed into representations that encode gait dynamics and movement patterns, which can be analyzed using image-based learning models.

The objective of this study is to investigate how artificial intelligence methods can leverage non-contact sensing data to distinguish between healthy individuals and Parkinson’s disease patients. Particular attention is given to the integration of multi-sensor signals, the extraction of informative representations of human motion, and the design of robust classification models evaluated on clinical datasets.

Overall, the proposed approach contributes to the development of scalable and non-invasive screening tools that could support both clinical assessments and future remote or domestic monitoring applications.